Computational Complexity

Reading guide and Homework for **Week 07**

1 Reading Guide

This week we have started studying combinatorial models of computation.

- We studied Boolean circuits and formulas. Formulas are not explicitly talked about in the book. But you can read §6.1 to learn about circuits.
- We learned that there exist functions that can't be computed by small circuits. §6.5

2 Homework

1. Show that, given a Boolean formula $\varphi(x_1, \ldots, x_n)$ of size (number of leaves) s, there exists an equivalent Boolean formula φ' of size $\leq s$ where all negations are at the bottom. In other words, φ' is a formula with only AND and OR gates, over the input $\{x_1, \bar{x}_1, \ldots, x_n, \bar{x}_n\}$ of variables and negated variables.

2. Show that the most $(n+1)$-th bit of the addition of two n-bit natural numbers can be computed by polynomial-size circuits.

   More precisely, consider the Boolean function: $\mathsf{ADD}_n : \{0,1\}^{2n} \to \{0,1\}^{n+1}$, which, given $2n$ bits as input, interprets these bits as encoding two natural numbers between 0 and $2^n - 1$. Then:

   $$\mathsf{ADD}_n(x, y) = x + y$$

   with the result encoded in binary using $n+1$ bits.

   Show that $\mathsf{S}(\mathsf{ADD}_n) = n^c$ for some constant c. How small can you make c? Hint: use the primary school addition algorithm (for bits instead of decimals). Begin by constructing circuits to determine if there is a carry at the i-th position. Hint 2: If stuck, read Section 1.1 of the book "Introduction to Circuit Complexity" by Heribert Vollmer.

3. Show that, in any binary (fan-in ≤ 2) Boolean formula $\varphi(x_1, \ldots, x_n)$ with $\mathsf{L}(\varphi) = S$ leaves, there exists a "balanced" node v having at least $\frac{1}{3}S$ and at most $\frac{2}{3}S$ descendant leaves.

4. Let $\varphi_v(x_1, \ldots, x_n)$ be the sub-formula computed at such a "balanced" node, and let $\psi_v(x_v, x_1, \ldots, x_n)$ be the formula obtained from φ be replacing the sub-formula at node v with an extra variable x_v. Make a drawing to understand what's going on. Then by the choice of v we have $\mathsf{L}(\varphi_v)$ and $\mathsf{L}(\psi_v)$ are both between $\frac{1}{3}S$ and $\frac{2}{3}S$.

   Observe that we can write φ as $\varphi(x_1, \ldots, x_n) = [\varphi_v(x_1, \ldots, x_n) \wedge \psi_v(1, x_1, \ldots, x_n)] \vee [\neg\varphi_v(x_1, \ldots, x_n) \wedge \psi_v(0, x_1, \ldots, x_n)]$.

   Explain how any formula φ of size $\leq S$ can be converted into a formula of depth $O(\log S)$. (Hint: Apply the above recursively.)

   Finally, using all of the above, show that for any Boolean function $F : \{0,1\}^n \to \{0,1\}$, $\mathsf{D}(F) = \Theta(\log \mathsf{L}(F))$.

5. Show that, given a Boolean *circuit* $\phi(x_1, \ldots, x_n)$ of size (number of leaves) s, there exists an equivalent Boolean circuit ϕ' of size $O(s)$ where all negations are at the bottom. In other words, ϕ' is a circuit with only AND and OR gates, over the input $\{x_1, \bar{x}_1, \ldots, x_n, \bar{x}_n\}$ of variables and negated variables. Hint: unlike in formulas where you can get away with roughly the same size, here you will need twice the size.

6. Consider the function *iterated Boolean matrix multiplication* $\mathsf{IMM}_{n,\ell} : (\{0,1\}^{n\times n})^\ell \to \{0,1\}^{n\times n}$, where the input is interpreted as ℓ-many $n \times n$ 0/1-matrices, and the output is

$$\mathsf{IMM}(A_1, \ldots, A_\ell) = A_1 \cdot \ldots \cdot A_\ell$$

where each $\cdot$ denotes the Boolean matrix product:

$$(A \cdot B)_{i,j} = \bigvee_{k=1}^{n} A_{ik} \wedge B_{kj}.$$

Show that $\mathsf{D}(\mathsf{IMM}_{n,\ell}) = O(\log \ell \cdot \log n)$, and $\mathsf{S}(\mathsf{IMM}_{n,\ell}) \leq \ell n^3$.

7. Let $n = 2^k$ be a power of 2. Define the *pointer jumping function* $F_{n,\ell} : \{0,1\}^{kn} \to \{0,1\}^k$ as follows. Interpret the input as a sequence $x = (x_0, x_1, \ldots, x_{n-1})$ of n blocks of $k = \log n$ bits each. Define:

$$y_0 = x_0 \qquad y_{i+1} = x_{y_i}$$

where by x_{y_i} we mean the block indexed by the value y_i (so we interpret y_i as a natural number between 0 and $n-1$, written in binary). Then $F_{n,\ell}(x) = y_\ell$.

Show that $\mathsf{S}(F_{n,\ell}) \leq \ell n \log n$, and $\mathsf{D}(F_{n,\ell}) \leq \ell \log n$. Show in fact that $\mathsf{D}(F_{n,\ell}) \leq \log \ell \cdot \log n$.